COMMUNICATIONS AND CODE-BREAKING EQUIPMENT OF WORLD WAR II

Between World War I and World War II great strides were made in both wired and wireless communications technology, the latter reaching a point at which it was possible to build reliable transmitter/receiver sets (transceivers) that would fit into a small suitcase. By 1939, mechanical calculators were commonplace in the business world, and the technology they employed was increasingly used to mechanize the encoding of messages, making them very difficult to decipher if intercepted. During World War II, these two apparently disparate disciplines joined forces to create the first electromechanical (and later electronic) computers, which were developed specifically as aids to code-breaking.

▼ ENIGMA

Date 1926–1950s

Origin Germany

Type Encoding/decoding device

The operator turned Enigma's rotors to a random setting and typed the message to be encoded using the keys below – each keystroke advancing the rotors so that repeating the same keystroke gave a different result each time. Decoding reversed this process.

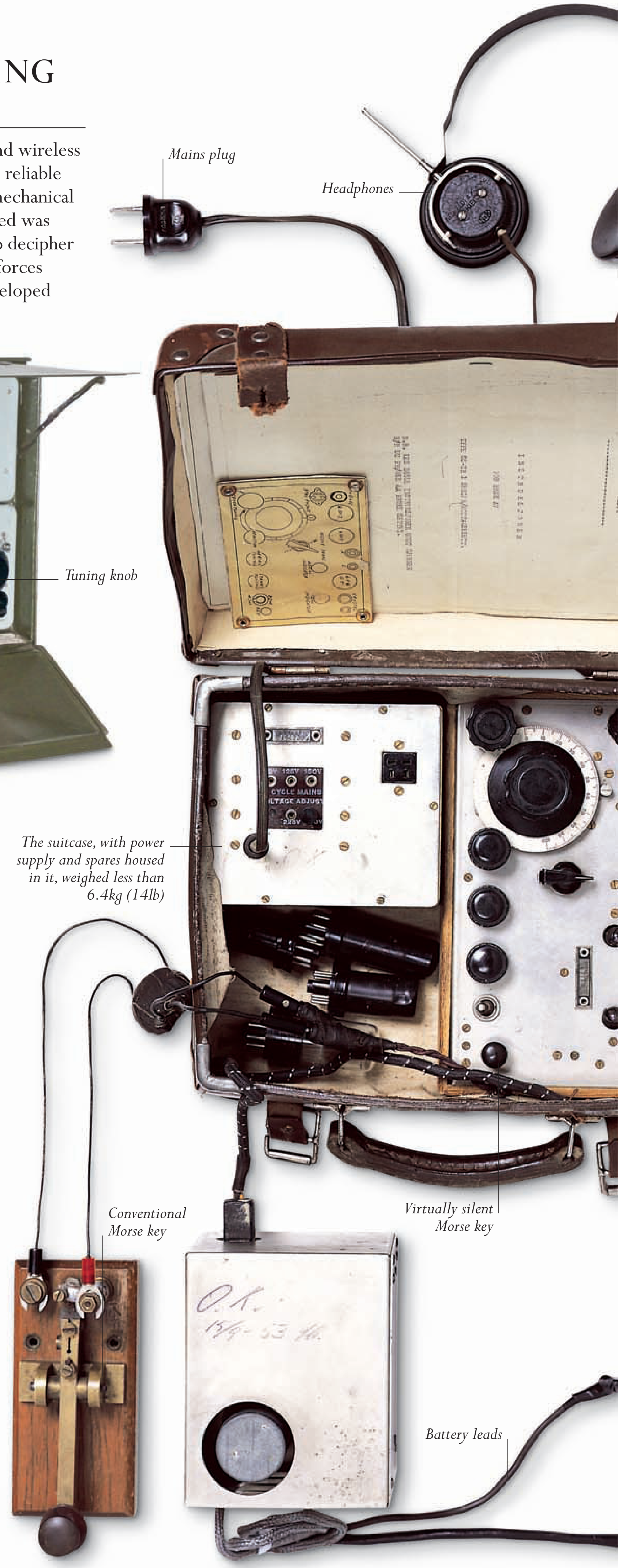
▲ WS18

Date 1939

Origin UK

Type Radio transceiver

Described as being "for short range telephony in forward areas", the Wireless Set 18 had an effective transmission range of 8km (5 miles). It was issued in 1939, and was the first "man-pack" radio transceiver put into series production for the British Army.



▶ PARASET SUITCASE RADIO TRANSCEIVER

Date 1940

Origin UK

Type Radio transceiver

The first miniature radio transceiver for clandestine use, the Whaddon Mk VII Paraset was the absolute minimum needed to set up two-way communication over distances of up to 800km (500 miles). It included a built-in Morse key that was almost silent in operation.